

WHITEMAN AFB, MO, USA

WMO: 724467

Lat: 38.717N

Lon: 93.550W

Elev: 265

StdP: 98.18

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13930

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.0	-12.9	-20.7	0.6	-14.8	-17.9	0.8	-11.4	12.8	4.6	11.6	2.7	4.0	330	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	35.9	24.7	33.9	24.4	32.5	24.2	26.7	31.9	25.9	31.2	25.3	30.4	5.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.4	21.3	28.8	24.6	20.2	28.5	23.8	19.3	28.1	85.0	31.9	81.7	31.2	78.6	30.5	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.7	8.7	DB	-19.8	37.5	3.3	1.9	-22.1	38.9	-24.0	40.0	-25.9	41.0	-28.3	42.3	(4)
(5)			WB	-20.0	28.0	3.1	0.9	-22.3	28.6	-24.1	29.1	-25.9	29.5	-28.1	30.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.3	-1.0	1.8	7.2	13.0	18.6	23.8	26.0	25.1	20.9	14.3	7.3	1.3	(6)	
	DBStd	10.69	6.69	6.75	6.41	5.16	4.39	3.28	3.21	3.16	4.34	5.13	5.70	6.06	(7)	
	HDD10.0	1150	344	239	128	28	0	0	0	0	18	116	275		(8)	
	HDD18.3	2663	599	462	346	172	53	3	0	1	23	145	333	527	(9)	
	CDD10.0	2338	3	11	43	118	267	414	497	467	327	151	34	6	(10)	
	CDD18.3	810	0	0	2	12	60	167	239	209	100	19	1	0	(11)	
	CDH23.3	7983	0	1	22	112	494	1601	2594	2075	913	163	7	0	(12)	
	CDH26.7	3348	0	0	1	17	132	643	1225	938	355	37	0	0	(13)	
(14) Wind	WSAvg	4.1	4.7	4.7	4.8	4.9	4.0	3.9	3.3	3.1	3.4	4.0	4.4	4.3	(14)	
(15) Precipitation	PrecAvg	1087	43	52	71	118	140	142	112	86	106	86	77	53	(15)	
	PrecMax	1517	138	139	154	337	322	299	388	199	323	239	266	130	(16)	
	PrecMin	675	7	5	28	24	34	31	11	22	17	11	16	7	(17)	
	PrecStd	202	32	33	35	55	69	72	82	48	76	52	59	30	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.9	22.2	25.9	28.8	32.0	35.0	38.5	37.8	35.2	30.0	24.1	19.9	(19)	
		MCWB	13.6	14.8	17.0	19.0	23.1	25.0	24.0	23.9	23.0	21.6	16.9	15.2	(20)	
	2%	DB	15.2	18.2	23.0	26.4	29.9	33.2	36.3	35.2	32.6	27.5	21.5	16.6	(21)	
		MCWB	10.6	12.3	15.5	17.5	21.8	24.5	25.1	24.4	23.1	18.8	15.0	13.0	(22)	
	5%	DB	12.1	15.4	20.5	24.2	28.0	32.1	34.3	33.5	30.8	25.0	18.8	12.9	(23)	
		MCWB	8.2	11.0	14.1	16.5	20.8	24.1	25.0	24.2	22.7	18.0	13.9	9.1	(24)	
	10%	DB	8.4	12.3	17.5	22.1	26.3	30.6	32.6	31.8	28.6	22.7	16.2	10.5	(25)	
		MCWB	5.3	8.5	11.7	15.5	19.7	23.4	24.8	23.8	21.4	17.1	12.0	7.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.9	16.6	19.1	20.8	24.4	26.9	27.7	27.8	25.7	23.5	18.3	16.8	(27)	
		MCDB	17.5	19.8	23.7	26.3	29.4	31.8	32.8	31.8	29.8	27.9	22.5	18.9	(28)	
	2%	WB	11.9	14.0	16.9	19.5	23.2	25.8	26.8	26.6	24.6	21.4	16.4	13.6	(29)	
		MCDB	14.2	17.0	21.0	24.1	27.8	31.0	32.4	31.3	29.4	24.3	19.8	16.2	(30)	
	5%	WB	8.4	11.3	14.9	18.1	22.1	25.1	26.1	25.7	23.7	19.8	14.7	10.3	(31)	
		MCDB	11.6	14.7	19.2	22.2	26.2	30.1	31.7	30.7	28.5	23.3	17.8	12.1	(32)	
	10%	WB	5.5	8.4	12.6	16.6	20.8	24.2	25.4	24.8	22.7	18.1	12.6	7.5	(33)	
		MCDB	8.3	12.4	16.8	20.9	24.8	29.1	30.8	29.8	27.0	21.7	15.8	10.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	10.3	11.4	11.5	10.8	10.5	10.6	11.1	11.8	11.7	10.8	9.4	(35)	
		MCDBR	14.8	15.0	15.3	14.6	12.3	11.9	12.4	13.2	13.2	13.7	14.0	13.3	(36)	
	5% WB	MCWBR	10.5	10.4	8.9	8.2	6.2	5.1	4.1	4.9	5.4	6.9	9.1	9.6	(37)	
		MCWBR	13.5	13.5	13.3	11.9	10.9	10.8	10.5	11.1	11.4	11.1	12.1	11.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.297	0.310	0.337	0.388	0.425	0.427	0.444	0.447	0.387	0.346	0.307	0.292	(40)	
		taud	2.463	2.427	2.410	2.281	2.238	2.265	2.234	2.234	2.359	2.461	2.514	2.499	(41)	
	Ebn at Noon		884	917	923	889	858	854	835	824	857	865	867	865	(42)	
		Edh at Noon	80	95	107	130	138	135	138	135	112	91	76	72	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.17	2.90	3.81	4.91	5.68	6.43	6.42	5.65	4.81	3.42	2.36	1.87	(44)	
	RadStd		0.25	0.34	0.34	0.32	0.51	0.52	0.40	0.32	0.31	0.36	0.25	0.18	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.74	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page